

Education

2021 – 2025, Ph.D Theoretical Physics, University of Leeds

Supervisors: Prof. Zlatko Papic, Prof. Jiannis Pachos.

Project Title: Controlling, Evading, and Maximising Quantum Thermalisation.

- Computationally focused project: Using python, simulated complex quantum systems and analysed/visualised datasets.
- 5 published papers (PRXQ, PRB, Scientific Reports).
- Presented research at academic conferences (CMD, IQTN, ETN, IOP, APS) and summer schools.
- Supervised 8 BSc/MSc student projects and visiting work experience students.
- Ran 5 “hands-on” quantum computing workshops for visiting students/schools.

2017 – 2021, MPhys & BSc Theoretical Physics, University of Leeds, 1st Class with Honours

- Achieved 1st Class in every single module – (Ranked top 1% out of approximately 200 students).

2011 – 2017, Leeds West Academy, Leeds

- A Levels – Mathematics (A*), Physics (A), Chemistry (A), Extended Learning Project (A).
- GCSEs – 12 GCSEs grade A*- C (Including Mathematics (A*) and English (A)).

Work History

2025 – Present, Postdoctoral Researcher, University of Leeds, School of Physics & Astronomy

- Researching quantum many-body systems through numerical modelling, data analysis and data visualisation in python.
- Frequent use of version control pipelines such as GitHub, bitbucket, and overleaf to share manuscripts and code.
- Coordinated with colleagues internationally in UK, US, Europe running simulations on high performance computing hardware in Leeds and Durham.

2022 – 2025, Teaching Assistant, University of Leeds, School of Physics & Astronomy

- Facilitated teaching the modules “Physics 2” and “Maths 1”: leading workshops in both quantum and condensed matter physics – answering questions and helping students garner a more in-depth understanding.
- Acted as a substitute for an absent lecturer and lectured to 100+ students.
- Spearheaded 50+ workshops: developed presenting skills, time management, project organising, and managing under pressure.

2021, Undergraduate Mentor, IOP Institute of Physics, Brightside

- Influenced a research trial over 3 months to improve gender balance in A-Level Physics.
- Mentored 3 year 12 students. Provided guidance with exam preparation/university.
- Credited with going beyond the scope of the programme by providing tutoring and sending thoughtful messages to all 3 students.

2019 – 2020, Personal school tutor, University of Leeds, School of Education

- Visited a local high school weekly and mentored a pupil premium student weekly with STEM.
- Influenced a global research project: “ASCENTS” facilitated 845 economically challenged students tackle GCSE Science. Students performed 30% better on average -- developed independent networking and travel skills.

2018, School representative/assistant, University of Leeds, School of Physics & Astronomy

- Managed a group of year 12 students from across the country. Touring students across campus while ensuring safety. Guided students with managing research projects.
- Hosted a poster session at the end to academics and alumni.
- Inspired students to attend the University and consider research as a job.
- Acted as a team with other employees to coordinate each day.

Additional Volunteering/Experience

2022 – 2024, A-Level maths/physics tutor, University of Leeds, School of Physics & Astronomy

- Dedicated 2 hours a week to volunteer for tutoring Maths/Physics to A-level students as part of the University of Leeds Theoretical Physics group public tutoring scheme.

2015 – 2017, Student ambassador, Leeds West Academy

- Volunteered as a STEM student ambassador for guests and aspiring students.
- Co-ordinated taster days, talking about STEM Subjects to both students and parents, demonstrating passion and knowledge for the subject areas.
- Aided the head of science in a yearly chemistry show in front of audiences.
- Demonstrated scientific experiments in front of an audience (ages from 12 to 60+).

Additional Skills

IT skills:

- Proficient in Linux OS, Windows OS, Python (particularly scientific/mathematical packages), LaTeX, Microsoft Office packages, Graphic design software, Adobe Premier Pro, Adobe Photoshop, Inkscape.
- Modelled protein molecular simulation using programs such as MD, Pymol, Chimera.
- Experience using the ARC supercomputer based in Leeds and BEDE supercomputer in Durham. Well versed in running Linux OS and Windows OS with HPC.
- Proficient using Quantum Computing suits and facilities through IBM's Qiskit.

Design skills:

- Graphic design skills and woodworking skills with associated qualifications.

Undergone safeguarding training/DBS Check

Awards

- Academic Fellow: Awarded Leeds Doctoral Scholarship due to academic achievements and presented research proposal. Allocated a fully funded PhD and maintenance grant. (Similarly, offered at York University). Fully funded 4 years of study with maintenance grant.
- Stoner Award: Awarded to the student on the Theoretical Physics programme with the highest classification average at the end of the degree (in a cohort of 200+ students).
- Award for Best Student 2019 in the University of Leeds 2nd Year Physics Program, achieving the highest average in a cohort of 300+ students.
- Granted hardest working student award out of 100 students in 2017 at Leeds West Academy.
- Won 2nd place in the Leeds Rotary engineering competition 2 years in a row.
- 1st to achieve an A* in A-Level Mathematics at Leeds West Academy.
- Won a competition for best graphically designed mousepad: subsequently commercially sold.

Interests and Achievements

- Co-ordinated in founding a university society: "DiceSoc", a Dungeons & Dragons society, fostering a culturally encouraging and social environment.
- Physics/Maths – Enjoy reading and learning more on the subject and related areas. Competed in maths competitions and researched leisurely.
- Travelling – Travelled to Norway, Sweden and America and have enjoyed seeing new cultures and environments. Working understanding of the Norwegian language.
- Writing – Enjoy reading into the fantasy genre and exploring different worlds through literature.